RETAIL BANKING · CREDIT RISK MANAGEMENT

Banking Loan Risk Analysis Application

Full-stack automated ETL, relational risk mart, explainable risk bands, and a Streamlit underwriting portal.

Python 3 · FastAPI · MySQL · Pandas · Streamlit · JWT RBAC

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Manual risk monitoring slows decisions and weakens auditability

Credit teams managing tens of thousands of card lines need a faster, more consistent way to see exposure, payment behavior, and emerging default risk.

Monitoring friction

Spreadsheet-heavy monitoring is slow, error-prone, and difficult to audit.

Exposure questions

Which customer segments carry materially higher default risk?

Behavior context

How do limits, balances, and payment history vary across demographics?

Action under policy

Which accounts need urgent manual review under policy rules?

How does monthly credit exposure compare with cash collections?

THE SOLUTION

The solution turns raw files into explainable underwriting action

Each layer removes ambiguity between incoming data and the decision a credit professional needs to make.

Automate

Ingest new files and eliminate manual spreadsheet wrangling.

Validate

Run 8 automated data-quality checks and reconcile records with MD5 checksums.

Explain

Assign transparent Low, Medium, and High risk bands instead of black-box decisions.

Operationalize

Give Underwriters review queues and What-If simulation, while Risk Directors get portfolio KPIs.

ONE OPERATING LOOP

One application connects ETL, scoring, review, and executive visibility



Ingest

Drop new files into `data/raw/`; process into MySQL and archive automatically to `data/processed/`.

Score

Score accounts on a 0–100 scale with explainable Low, Medium, and High bands.

Review

Route rule-triggered accounts to a manual review queue for approve / decline decisions.



See the portfolio

FastAPI exposes paginated, sorted, JWT-protected endpoints; Streamlit tracks KPIs, cashflow trends, and live scenarios.

ARCHITECTURE

Six layers move 30,000 customer records from source to decision

Data flows into governed decisions and review.

DATA PATH

Source

UCI Credit Card Clients Benchmark Dataset with 30,000 customers

DATA PATH

Ingestion

Pandas cleaning, 8 validation rules, and MD5 reconciliation

DATA PATH

Storage

Normalized 3NF MySQL across customers, credit profiles, bills, payments, and indexed segment summaries

DECISION & REVIEW

Rules

Utilization, repayment, delinquency delay, and exception triggers

DECISION & REVIEW

API

FastAPI with OAuth2/JWT authentication and Analyst vs Administrator RBAC

DECISION & REVIEW

UI

Streamlit multi-tab dashboard with Plotly charts and What-If simulation

TECHNICAL FOUNDATION

The stack is optimized for speed, integrity, and explainability

The business value is deliberate control: fast access, trusted records, and decisions that can be defended.

FastAPI / Uvicorn

Asynchronous ASGI performance, automatic OpenAPI documentation, and strict Pydantic V2 schemas.

MySQL

ACID storage, foreign keys, connection pooling, and compound indexes for fast queries.

Pandas

Vectorized ETL, categorical decoding, and derived features such as age groups and ratios.

Streamlit / Plotly

Responsive analytics, live sliders, and visual KPI meters.

PyJWT / Cryptography

Token-based role control for enterprise-style access management.

DECISION LOGIC

Risk bands make every recommendation explainable

The score is a compact signal. The thresholds and triggers show exactly why an account received it.

Low · < 35

Max delay ≤ 0, utilization < 60%, and repayment ratio ≥ 50%

Medium · 35–64

Max delay = 1 month, utilization 60%–85%, or repayment ratio 20%–50%

High · 65+

Max delay ≥ 2 months, or utilization ≥ 85% with repayment < 20%, or historical default

Manual review triggers

- High exposure with delay
- Utilization > 90% with repayment < 15%
- Chronic delinquency across 2+ cycles
- Statement > NT$50,000 with zero recent payment

UNDERWRITING PORTAL

The portal turns portfolio analytics into daily underwriting workflow

One secure surface connects portfolio context to the next account-level decision.

Secure login

JWT sign-in separates Administrator and Analyst access.

Executive KPIs

Track exposure, average credit limit, default rate, and risk-band mix.

Customer ledger

Search by Customer ID; compare six-month statements with cash payments.

Review queue

Inspect triggered rules and record approve / reject decisions.

What-If simulator

Test applicant scenarios with a live risk gauge and recommendation.

ETL batch center

Upload new files with automatic archival.

BENCHMARKS AND INSIGHTS

Benchmarks prove reliability and actionable insights

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AUTOMATED TESTS

Pytest cases passed across ETL, business rules, API, and database operations.

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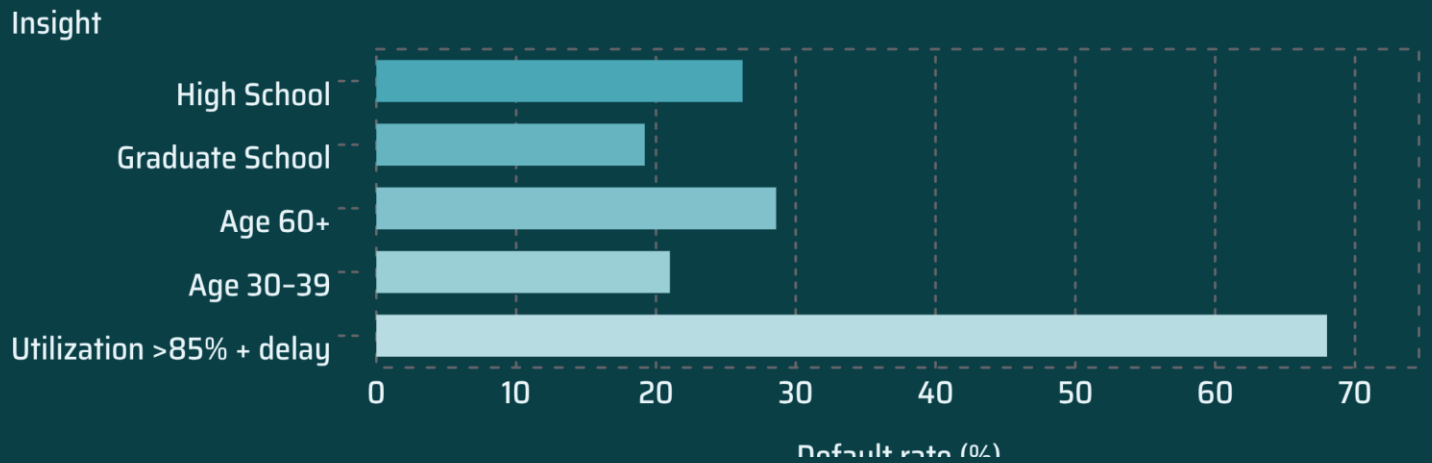
3,000 RECORDS

Ingested with 100% MD5 reconciliation.

<25 ms

API RESPONSE

Paginated query response time.



WHAT THE DATA SAYS

Default risk concentrates in older, high-utilization, and delayed-payment segments.

Education: High School 26.2% vs Graduate School 19.2%. Age 60+: 28.6% vs 21.0% for ages 30–39. Above 85% utilization plus payment delay drives default above 68%.

THE PATH FORWARD

From capstone prototype to production risk platform

Reliable data foundations first, explainable decisions today, predictive risk intelligence next.

Normalize the model

Replace 18 repetitive monthly columns with indexed one-to-many bills and payments records.

Scale the platform

Refresh pre-aggregated segment KPI summaries during ETL, then containerize with Docker for AWS ECS or Azure Container Apps.

Extend intelligence

Add Apache Kafka for real-time transaction streaming and complement deterministic rules with XGBoost probability scores.

A trusted pipeline turns raw lending data into a decision that teams can act on, explain, and improve.